5.3

1) Find reference angle

(a) $205°$

Q III

$\bar{\theta} = 205° - 180°$

$= \boxed{25°}$

(b) $830°$

$\frac{830°}{360°} \rightarrow$ remainder $110°$

Q II

$\bar{\theta} = 180° - 110°$

$= \boxed{70°}$

(c) $-95°$

Q III

$\bar{\theta} = 180° - 95°$

$= \boxed{85°}$

2) Find reference angle

(a) $\frac{7\pi}{9}$

Q II

$\bar{\theta} = \pi - \frac{7\pi}{9}$

$= \boxed{\frac{2\pi}{9}}$

(b) -0.4π

coterminal angle: 1.6

Q IV

$\bar{\theta} = 2\pi - 1.6\pi$

$= \boxed{0.4\pi}$

(c) 0.4

Q I

$\bar{\theta} = \boxed{0.4}$

3) Find exact value of trig function

$\sin\left(\frac{7\pi}{2}\right)$

$= \sin\left(\frac{3\pi}{2}\right)$

$= \boxed{-1}$

4) Find exact value of trig function

$\cos\left(\frac{5\pi}{6}\right)$

Q II

reference angle:

$\pi - \frac{5\pi}{6} = \frac{\pi}{6}$

$-\cos\left(\frac{\pi}{6}\right)$

$= \boxed{-\frac{\sqrt{3}}{2}}$

5) write first trig function in terms of the second for θ in given quadrant

$\tan(\theta), \cos(\theta)$; quadrant IV

$\tan(\theta) = ?$

$\sin^2\theta + \cos^2\theta = 1$

$\sin\theta = -\sqrt{1-\cos^2\theta}$

$\tan\theta = \frac{\sin\theta}{\cos\theta}$

$\boxed{\tan\theta = \frac{-\sqrt{1-\cos^2\theta}}{\cos\theta}}$

6) write first trig function in terms of the second for θ in given quadrant

$\cos\theta$, $\sin\theta$; quadrant III

$\cos\theta = ?$

$\sin^2\theta + \cos^2\theta = 1$

$\cos^2\theta = 1 - \sin^2\theta$

$\cos\theta = -\sqrt{1-\sin^2\theta}$

7) Find values of trig functions of θ

$\cos(\theta) = \frac{6}{17}, \sin\theta < 0$

$\sin^2\theta + \cos^2\theta = 1$

$\sin^2\theta + \left(\frac{6}{17}\right)^2 = 1$

$\sin^2\theta + \frac{36}{289} = 1$

$\sin^2\theta = \frac{253}{289}$

$\sin\theta = \boxed{-\frac{\sqrt{253}}{17}}$

$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-\frac{\sqrt{253}}{17}}{\frac{6}{17}} = \boxed{-\frac{\sqrt{253}}{6}}$

$\csc\theta = \frac{1}{\sin\theta} = \frac{1}{-\frac{\sqrt{253}}{17}} = -\frac{17}{\sqrt{253}} = \boxed{\frac{-17\sqrt{253}}{253}}$

$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\frac{6}{17}} = \boxed{\frac{17}{6}}$

$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{-\frac{\sqrt{253}}{6}} = -\frac{6}{\sqrt{253}} = \boxed{-\frac{6\sqrt{253}}{253}}$

8) Find values of trig functions of θ

$\csc\theta = 4$, θ in QI

$\csc\theta = \frac{1}{\sin\theta}$

$\boxed{\sin\theta = \frac{1}{4}}$

$\left(\frac{1}{4}\right)^2 + \cos^2\theta = 1$

$\frac{1}{16} + \cos^2\theta = 1$

$\cos^2\theta = \frac{15}{16}$

$\boxed{\cos\theta = \frac{\sqrt{15}}{4}}$

$\tan\theta = \frac{\sin\theta}{\cos\theta}$

$= \frac{\frac{1}{4}}{\frac{\sqrt{15}}{4}}$

$= \frac{1}{\sqrt{15}}$

$\boxed{\tan\theta = \frac{\sqrt{15}}{15}}$

$\sec\theta = \frac{1}{\cos\theta}$

$= \frac{4}{\sqrt{15}}$

$\boxed{\sec\theta = \frac{4\sqrt{15}}{15}}$

$\cot\theta = \frac{1}{\tan\theta}$

$= \frac{1}{\frac{1}{\sqrt{15}}}$

$\boxed{\cot\theta = \sqrt{15}}$

9) if $\theta = \frac{\pi}{4}$, find value of each expression

$\sin^2\theta = ?$

$\sin(\theta^2) = ?$

$\sin^2\theta = \sin^2\left(\frac{\pi}{4}\right)$

$= \left(\frac{\sqrt{2}}{2}\right)^2$

$\boxed{\sin^2\theta = \frac{1}{2}}$

$\sin(\theta^2) = \sin\left(\left(\frac{\pi}{4}\right)^2\right)$

$= \sin\left(\frac{\pi^2}{16}\right)$

$\boxed{\sin(\theta^2) \approx 0.578}$

10) find area of a triangle given description – a triangle with sides of length 15 and 23 and included angle 10°

$A = \frac{1}{2} ab \sin\theta$

$= \frac{1}{2} \cdot 15 \cdot 23 \cdot \sin(10°)$

$\boxed{\approx 30 \text{ un}^2}$

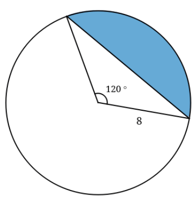
11) find area of a triangle given description – an equilateral triangle with side of length 10

$A = \frac{1}{2} ab \sin(60°)$

$= \frac{1}{2} \cdot 10 \cdot 10 \cdot \sin(60°)$

$\boxed{\approx 43.3 \text{ un}^2}$

12) find area of shaded region



Area of sector (blue part and triangle)

$\frac{120°}{360°} \cdot \pi(8^2)$

Area of triangle

$\frac{1}{2} 8^2 \sin 120°$

Area of shaded region

$\frac{120°}{360°} \cdot \pi(8^2) - \left(\frac{1}{2} 8^2 \sin 120°\right)$

$\boxed{\approx 39.3 \text{ un}^2}$

13) find exact value of trig function

$\sec(-405°)$

$\sec(-405°) = \sec(315°)$

Q IV

$\bar{\theta} = 360° - 315° = 45°$

$\sec(45°)$

$= \boxed{\sqrt{2}}$

14) find exact value of trig function

$\cot(585°)$

$\cot(585°) = \cot(225°)$

Q III

$\bar{\theta} = 225° - 180° = 45°$

$\cot(45°) = \boxed{1}$

15) find exact value of trig function

$\csc\left(\frac{2\pi}{3}\right) \rightarrow$ Q II

$\csc\theta = \frac{1}{\sin\theta}$

$= \frac{1}{\sin\left(\frac{2\pi}{3}\right)}$

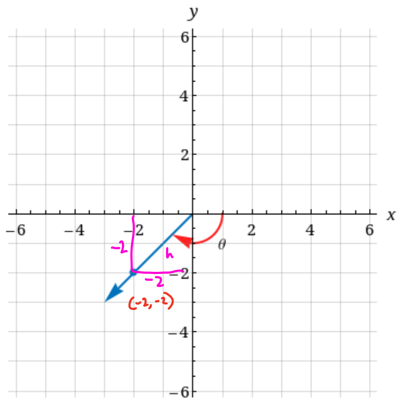
$= \frac{1}{\sin\left(\frac{\pi}{3}\right)}$

$= \frac{1}{\frac{\sqrt{3}}{2}}$

$= \frac{2}{\sqrt{3}}$

$= \boxed{\frac{2\sqrt{3}}{3}}$

16) angle is given in coordinate plane



find exact value of $\cos\theta$.

$h = \sqrt{(-2)^2+(-2)^2}$

$= 2\sqrt{2}$

$\cos\theta = \frac{-2}{2\sqrt{2}}$

$= -\frac{\sqrt{2}}{2}$

17) find values of trig functions of θ

$\sin\theta = \frac{6}{11}$, $\sec\theta > 0$

Q I

$\sin^2\theta + \cos^2\theta = 1$

$\left(\frac{6}{11}\right)^2 + \cos^2\theta = 1$

$\boxed{\cos^2\theta = \frac{\sqrt{85}}{11}}$

$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\frac{6}{11}}{\frac{\sqrt{85}}{11}} = \frac{6}{\sqrt{85}} = \boxed{\frac{6\sqrt{85}}{85}}$

$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\frac{\sqrt{85}}{11}} = \frac{11}{\sqrt{85}} = \boxed{\frac{11\sqrt{85}}{85}}$

$\csc\theta = \frac{1}{\sin\theta} = \frac{1}{\frac{6}{11}} = \boxed{\frac{11}{6}}$

$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{\frac{6}{\sqrt{85}}} = \boxed{\frac{\sqrt{85}}{6}}$

18) write first trig function in terms of the second for θ in given quadrant

$\sec\theta$, $\tan\theta$; quadrant IV

$\sec\theta = ?$

$\tan^2\theta + 1 = \sec^2\theta$

$\sec\theta = \boxed{\sqrt{\tan^2\theta + 1}}$

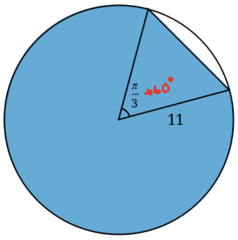
19) find area of a triangle given description – a triangle with sides of length 5 and 9 and included angle 27°

$A = \frac{1}{2}ab\sin\theta$

$= \frac{1}{2}\cdot 5\cdot 9\cdot \sin(72^\circ)$

A $\boxed{\approx 21.4 \text{ un}^2}$

20) find area of shaded region



Area of whole circle:

πr^2

$= \pi \cdot 11^2$

$= 121\pi$

area of white gap is area of sector − area of triangle

sector area:

$\frac{1}{2} r^2 \theta$

$= \frac{1}{2} \cdot 11^2 \cdot \frac{\pi}{3}$

area of triangle:

$\frac{1}{2} ab \sin(\theta)$

$= \frac{1}{2} \cdot 11 \cdot 11 \cdot \sin(60°)$

Area of shaded region is

whole circle area − (sector area − area of triangle)

$= 121\pi - \frac{1}{2} \cdot 11^2 \cdot \frac{\pi}{3} + \frac{1}{2} \cdot 11 \cdot 11 \cdot \sin(60°)$

$\boxed{\approx 369.2 \text{ un}^2}$